

Gemini Evaluation Prompt

RAG Chatbot vs LM Studio — AI Judge Instructions

How to Use This Document

This document contains everything you need to evaluate two AI systems using Gemini as a neutral judge. Follow the steps below, then paste the prompt (Page 2) into Google AI Studio along with the three uploaded files.

Step 1	Go to aistudio.google.com and start a new chat with Gemini 1.5 Pro or Gemini 2.0.
Step 2	Upload these three files: <code>ai_evaluation_test_document.pdf</code> <code>evaluation_dataset.csv</code> <code>your_answers.csv</code>
Step 3	Copy the full prompt from Page 2 of this document and paste it into the chat.
Step 4	Let Gemini read all files and generate scores. It may take 1-2 minutes.
Step 5	Copy Gemini's output into a spreadsheet or share as a report.

What Files to Prepare

Before going to Gemini, you need to prepare one additional file — your answers CSV. This file should contain the answers collected from both systems for all 25 questions.

File	Description	Status
<code>ai_evaluation_test_document.pdf</code>	The 4-page source document both systems read	Ready — already generated
<code>evaluation_dataset.csv</code>	25 questions + ground truth answers	Ready — already generated
<code>your_answers.csv</code>	Answers from RAG Chatbot & LM Studio	You need to fill this in

Format for your_answers.csv

Create a CSV file with these exact columns and fill in one row per question:

```
id | question | ground_truth | rag_answer | lmstudio_answer
```

Note: Copy the `id`, `question`, and `ground_truth` columns directly from `evaluation_dataset.csv`. Then add `rag_answer` and `lmstudio_answer` columns with the responses you collected from each system.

The Gemini Prompt

Copy everything inside the box below and paste it into Gemini:

You are an expert, neutral AI evaluator. I have uploaded three files:

1. ai_evaluation_test_document.pdf – a 4-page report on AI in Healthcare.

This is the source document that both AI systems used to answer questions.

2. evaluation_dataset.csv – contains 25 test questions and their ground truth answers derived directly from the PDF.

3. your_answers.csv – contains answers from two AI systems for all 25 questions:

- System A: RAG Chatbot
- System B: LM Studio

YOUR TASK:

Evaluate BOTH systems on every question using the 4 metrics below.

Base your evaluation on the source PDF and the ground truth answers provided.

SCORING RUBRIC (score each metric 1 to 5):

Faithfulness (1-5): Does the answer contain ONLY information from the PDF?

5 = Fully grounded, no outside info added

3 = Mostly correct but adds minor unsupported claims

1 = Heavily relies on information not in the document

Relevancy (1-5): Does the answer directly address what was asked?

5 = Directly and fully addresses the question

3 = Partially addresses it but drifts off-topic

1 = Does not address the question at all

Correctness (1-5): How closely does the answer match the ground truth?

5 = Matches ground truth completely including all key facts/numbers

3 = Gets the main idea right but misses key details

1 = Factually wrong or contradicts the ground truth

Completeness (1-5): Does the answer cover all important points in the ground truth?

5 = All key points covered

3 = Covers main points but misses some details

1 = Missing most key information

SPECIAL RULES:

- For OUT-OF-CONTEXT questions (Q24, Q25): the correct answer is to say the information is not in the document. Give Faithfulness=5 if the system correctly says 'not found'. Give Faithfulness=1 if it invents an answer (hallucination).
- For SUMMARIZATION questions (Q19, Q20, Q21): give partial credit for each correct point covered.

OUTPUT FORMAT – respond EXACTLY in this structure for each question:

[Q01] Question text here

Difficulty: easy | Type: factual | Page: 1

System A (RAG Chatbot):

Faithfulness: X/5 – reason

Relevancy: X/5 – reason

Correctness: X/5 – reason

Completeness: X/5 – reason

Average: X.X/5

System B (LM Studio):

Faithfulness: X/5 – reason

Relevancy: X/5 – reason

Correctness: X/5 – reason

Completeness: X/5 – reason

Average: X.X/5

Winner: [System A / System B / Tie] – one sentence reason

After scoring ALL 25 questions, output this summary:

OVERALL SUMMARY TABLE

Metric	System A (RAG)	System B (LM Studio)
Faithfulness	X.XX	X.XX
Relevancy	X.XX	X.XX
Correctness	X.XX	X.XX
Completeness	X.XX	X.XX
OVERALL AVG	X.XX	X.XX

BREAKDOWN BY DIFFICULTY

Difficulty	System A (RAG)	System B (LM Studio)
Easy	X.XX	X.XX
Medium	X.XX	X.XX
Hard	X.XX	X.XX

BREAKDOWN BY QUESTION TYPE

Type	System A (RAG)	System B (LM Studio)
Factual	X.XX	X.XX
Summarization	X.XX	X.XX
Conceptual	X.XX	X.XX
Multi-hop	X.XX	X.XX
Out-of-context	X.XX	X.XX

FINAL VERDICT

Overall Winner: [System A / System B / Tie]

System A (RAG Chatbot) – Strengths:

- [point 1]
- [point 2]
- [point 3]

System A (RAG Chatbot) – Weaknesses:

- [point 1]
- [point 2]

System B (LM Studio) – Strengths:

- [point 1]
- [point 2]
- [point 3]

System B (LM Studio) – Weaknesses:

- [point 1]
- [point 2]

Hallucination Analysis:

System A hallucinated on: [list question IDs or 'none']

System B hallucinated on: [list question IDs or 'none']

Final Recommendation:

[2-3 sentences on which system to use, when, and why]

Evaluation Dataset — All 25 Questions

This is the full ground truth reference. These questions and answers are already in evaluation_dataset.csv which you will upload to Gemini. This page is for your own reference.

Q01 — What was the global investment in healthcare AI in 2023?

Ground Truth: Global investment in healthcare AI reached 45 billion USD in 2023, representing a 32% increase compared to the previous year.

Difficulty: easy | Type: factual

Q02 — What are the three core domains of AI in healthcare covered in this report?

Ground Truth: The three core domains are: (1) diagnostic assistance, (2) administrative automation, and (3) personalized treatment planning.

Difficulty: easy | Type: factual

Q03 — What accuracy rate did AI achieve in disease detection?

Ground Truth: Machine learning models demonstrated accuracy rates exceeding 94% in detecting diseases such as cancer, diabetic retinopathy, and cardiovascular disorders.

Difficulty: easy | Type: factual

Q04 — What percentage of surveyed hospitals adopted AI diagnostic tools?

Ground Truth: 67% of the surveyed hospitals adopted AI diagnostic tools.

Difficulty: easy | Type: factual

Q05 — By how much did AI reduce radiologist workload?

Ground Truth: AI-assisted imaging analysis reduced radiologist workload by an average of 38%.

Difficulty: easy | Type: factual

Q06 — How much did AI reduce administrative costs in pilot programs?

Ground Truth: AI-based automation tools reduced administrative costs by an average of 22% in pilot programs.

Difficulty: easy | Type: factual

Q07 — What was the NLP accuracy rate in data extraction tasks?

Ground Truth: Natural Language Processing (NLP) systems achieved a 97% accuracy rate in data extraction tasks.

Difficulty: easy | Type: factual

Q08 — What improvement in 5-year survival rates was seen in AI-assisted cancer treatment?

Ground Truth: Patients receiving AI-assisted treatment recommendations showed a 28% improvement in 5-year survival rates.

Difficulty: easy | Type: factual

Q09 — What percentage of respondents cited data privacy as a challenge?

Ground Truth: Data privacy was mentioned by 78% of respondents as the most frequently cited challenge.

Difficulty: easy | Type: factual

Q10 — What is the average cost of EHR-AI integration per facility?

Ground Truth: The average cost of EHR-AI integration was estimated at 1.2 million USD per facility.

Difficulty: easy | Type: factual

Q11 — What percentage of physicians trusted AI diagnostic recommendations without human verification?

Ground Truth: Only 41% of physicians surveyed expressed confidence in AI diagnostic recommendations without human verification.

Difficulty: easy | Type: factual

Q12 — What is the average upfront investment for a comprehensive AI system in healthcare?

Ground Truth: The average upfront investment was 3.7 million USD, with ongoing annual maintenance costs of approximately 480,000 USD.

Difficulty: easy | Type: factual

Q13 — How did AI affect the time to diagnosis in hospitals that adopted it?

Ground Truth: The average time to diagnosis decreased from 4.2 days to 1.8 days in facilities using AI tools.

Difficulty: medium | Type: factual

Q14 — What regulations were mentioned as creating data privacy challenges for AI in healthcare?

Ground Truth: HIPAA in the United States and GDPR in Europe impose strict requirements on how patient data can be used for AI training.

Difficulty: medium | Type: factual

Q15 — What is the 'black box' problem mentioned in the report?

Ground Truth: The 'black box' problem refers to the lack of explainability in AI decisions, raising concerns about trust and liability.

Difficulty: medium | Type: conceptual

Q16 — What percentage of rural hospitals had budget for AI adoption in the next 3 years?

Ground Truth: Only 29% of rural hospitals indicated they had sufficient budget to consider AI adoption within the next 3 years.

Difficulty: medium | Type: factual

Q17 — What interoperability standard does the report recommend for EHR integration?

Ground Truth: The report recommends HL7 FHIR to ensure new AI systems can connect seamlessly with existing EHR platforms.

Difficulty: medium | Type: factual

Q18 — How much can AI literacy programs increase physician trust in AI tools?

Ground Truth: AI literacy programs have been shown to increase physician trust in AI tools by up to 55% in pilot studies.

Difficulty: medium | Type: factual

Q19 — Summarize the key findings of the report across all three domains.

Ground Truth: 67% of hospitals adopted AI diagnostics (38% workload reduction, time-to-diagnosis from 4.2 to 1.8 days). Admin automation cut costs 22% with 97% NLP accuracy. AI treatment plans improved 5-year cancer survival by 28%.

Difficulty: hard | Type: summarization

Q20 — What are the main challenges to AI adoption in healthcare?

Ground Truth: Four challenges: (1) data privacy (HIPAA/GDPR, 78% cited), (2) legacy EHR integration (64% struggled, \$1.2M cost), (3) clinician trust (only 41% confident), (4) high costs (\$3.7M upfront).

Difficulty: hard | Type: summarization

Q21 — What does the report recommend for improving AI adoption in healthcare?

Ground Truth: Five recommendations: federated learning, staff AI training (+55% trust), HL7 FHIR standards, tiered funding for rural hospitals, and explainable AI (XAI) regulatory requirements.

Difficulty: hard | Type: summarization

Q22 — How does AI adoption differ between large hospitals and rural facilities?

Ground Truth: Large hospitals have higher adoption while only 29% of rural hospitals have budget for AI. Report recommends tiered funding programs with grants for smaller facilities.

Difficulty: hard | Type: multi_hop

Q23 — What is the potential annual cost saving from AI in global healthcare by 2030?

Ground Truth: AI has the potential to reduce global healthcare costs by an estimated 150 billion USD annually by 2030.

Difficulty: medium | Type: factual

Q24 — Which specific AI companies are mentioned in the report?

Ground Truth: The document does not mention any specific AI companies by name. [Correct answer = 'not found']

Difficulty: hard | Type: out_of_context

Q25 — What is the author's name of this report?

Ground Truth: The document does not mention any author name. [Correct answer = 'not found']

Difficulty: hard | Type: out_of_context

Scoring Tips & How to Interpret Results

Score Interpretation

Average Score	Interpretation	What to Do
4.5 – 5.0	Excellent — production ready	Deploy with confidence
3.5 – 4.4	Good — minor improvements needed	Fine-tune specific question types
2.5 – 3.4	Acceptable — clear gaps exist	Improve retrieval or prompting
1.5 – 2.4	Poor — significant issues	Rethink architecture or model
Below 1.5	Failing — not suitable for use	Major rebuild required

What Each Metric Reveals About Your System

Metric	If RAG scores higher	If LM Studio scores higher
Faithfulness	Your retrieval grounds answers well	LM Studio context injection is clean
Relevancy	Your prompting is focused	LM Studio stays on topic better
Correctness	Your RAG finds the right chunks	LM Studio covers full doc better
Completeness	Your chunks capture full context	Full-doc context helps completeness

Hallucination Watch — Out-of-Context Questions

Questions Q24 and Q25 have no answer in the PDF. The correct response for both systems is to say the information is not available in the document. If either system invents an answer (e.g. names a company or author that is not in the PDF), that is a hallucination and should receive a Faithfulness score of 1. This is one of the most critical signals for production readiness.

Tips for Reliable Gemini Scoring

Tip 1: Use Gemini 1.5 Pro or Gemini 2.0 Flash for best long-context reasoning over multiple files.

Tip 2: Run the evaluation twice and average scores if you want higher confidence in the results.

Tip 3: If Gemini skips a question, ask it to complete the missing ones in a follow-up message.

Tip 4: Ask Gemini to output scores as a JSON table at the end for easy copy-paste into a spreadsheet.

Tip 5: Do not tell Gemini which system is 'yours' — keeping it blind ensures unbiased scoring.

Good luck with your evaluation!